Forensics CLI 101 — Instructor Guide

Every challenge, answer, and accepted variant · generated 2026-08-18 from the live pack data

Quick reference
Set your own site URL and class password before handing this out. Admin handle: `warden`.
Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**.
30 challenges · 885 points total · flags look like `FLAG{ABCD2345EFGH}` and are **different for every student**.

How the difficulty fades — Act I briefs show the whole command, and the free hint repeats it. From Act II the brief names the tool but not its flags or arguments, and the exact line moves into a hint that costs XP. By the last act the brief states only the objective. A student who never buys a hint has genuinely recalled every command. Hint spend is visible in the instructor dashboard, so it reads as a difficulty signal rather than as cheating.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to **click the green flag in the terminal** to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. `cd ~` returns home.
"It says command not found"	If it's a real command we don't simulate (`top`, `nano`), the game says so explicitly. Genuine typos say "command not found".
"I don't know which command to use"	Intended from Act II on. Point them at the free hint first, then the Reference tab. The costed hint is the safety net, not a failure.
"The act is locked"	Each act opens after solving all but one of the previous act's challenges.
"Nothing printed"	Correct for `cd` and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: First Steps	Basic navigation, paths, prompt, and directory structure	7	90	Always open
Act II: Reading the Evidence	Inspecting files, headers, and forensic checksums	6	120	Solve 6 of 7 in the prior act
Act III: Search & Discovery	Pattern search, find, manual pages, and the WSL /mnt/c bridge	7	235	Solve 5 of 6 in the prior act

Act IV: The Plumbing	Pipes, redirection, filters, and multi-stage analysis	4	180	Solve 6 of 7 in the prior act
Act V: The Capstone	Multi-step investigation: partition offsets and evidence carving	2	180	Solve 3 of 4 in the prior act
Topside (Windows Quest)	Windows CMD parity: dir /a, type, findstr, attrib, and certutil	4	80	Always open

Students may skip **one** challenge per act and still advance.

Act I: First Steps

Basic navigation, paths, prompt, and directory structure

Where Am I?

10 pts · act1-pwd · OK

Task: Use the `pwd` command to print the directory you are standing in.

Answer: `pwd`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Type `pwd` (Present Working Directory) and press Enter.

Also accepted: `/bin/pwd`

Survey the Ground

10 pts · act1-ls · OK

Task: Use `ls` to list the files and directories where you are standing.

Answer: `ls`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Type `ls` to list files and folders.

Also accepted: `ls .`, `ls ./`

Hidden in Plain Sight

15 pts · act1-hidden · OK

Task: Something is hiding in your home directory. In Linux, files beginning with `.` are invisible to a plain `ls`. Use `ls -la` to find `.stash`, then read it with `cat .stash`. Submit the captured flag.

Answer: `ls -la` **then** `ls -a` **then** `cat .stash`

Completion: Submits a flag · flag is inside `/home/analyst/.stash`

Hint 1 (free): Run `ls -la` or `ls -a` to reveal files starting with a dot (`.`).

Hint 2 (–5 XP): Run `cat .stash` to read the hidden file, then submit the recovered flag.

One Level Down

15 pts · act1-cd · OK

Task: Move into `training/level_1`. Read `checkpoint_alpha.txt` and submit the recovered flag.

Answer: `cd training/level_1` **then** `cat checkpoint_alpha.txt`

Completion: Submits a flag · flag is inside `/home/analyst/training/level_1/checkpoint_alpha.txt`

Hint 1 (free): Use `cd training/level_1` to move down into it.

Hint 2 (–5 XP): Once inside level_1, run `cat checkpoint_alpha.txt` to view the flag.

Sideways Move

15 pts · act1-paths · OK

Task: Now reach `training/level_2` — a SIBLING of level_1, not a child. If you are inside level_1, you must go up before you can go over. Read `checkpoint_beta.txt` and submit the flag.

Answer: `cd training/level_2` **then** `cat checkpoint_beta.txt`

Completion: Submits a flag · flag is inside `/home/analyst/training/level_2/checkpoint_beta.txt`

Hint 1 (free): A sibling directory is reached by going UP first, then over. From your home directory the whole trip is one command: `cd training/level_2`.

Hint 2 (–5 XP): Once you are inside, run `cat checkpoint_beta.txt` and submit the flag.

Instructor note: From inside level_1 the sibling route is: cd .. then cd level_2 — that is the lesson.

Warp Speed (Tab Completion)

15 pts · act1-tab · OK

Task: Typing full directory paths is slow. Type `cd Doc` and press Tab to auto-complete `cd Documents`, then press Enter.

Answer: `cd Doc` **then** `cd Documents`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Type `cd Doc` then press the Tab key. The shell will auto-complete to `cd Documents`.

Also accepted: `cd ./Documents`, `cd ~/Documents`, `cd /home/analyst/Documents`

Instructor note: Students type cd Doc and press Tab. Typing it in full also counts.

Recall the Past (History)

10 pts · act1-history · OK

Task: Your recent commands are saved. Press the Up Arrow key to bring back a command you ran earlier — like `pwd` — then press Enter.

Answer: `pwd`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Run `pwd`, then press Up Arrow and Enter.

Instructor note: The lesson is the Up Arrow key, not the pwd command itself.

Act II: Reading the Evidence

Inspecting files, headers, and forensic checksums

Read the Case File

15 pts · act2-cat · OK

Task: Every investigation starts by opening the case file. Display `Documents/case_notes.txt` on screen using `cat`.

Answer: `cat Documents/case_notes.txt`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `cat` takes the path to the file you want printed.

Hint 2 (–5 XP): Run `cat Documents/case_notes.txt`.

Also accepted: `cat ./Documents/case_notes.txt`, `cat /home/analyst/Documents/case_notes.txt`

The Earliest Anomaly

15 pts · act2-head · OK

Task: `Documents/access.log` holds a long record. Show only its first 5 entries using `head`.

Answer: `head -n 5 Documents/access.log`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `head` prints the top of a file. One flag sets how many lines.

Hint 2 (–5 XP): Run `head -n 5 Documents/access.log`.

Also accepted: `head -n 5 ./Documents/access.log`, `head -5 Documents/access.log`

The Sensor Freeze

20 pts · act2-tail · OK

Task: The incident happened right before the sensors went dark, so the evidence is at the END of `Documents/access.log`. Read the final entries and submit the flag.

Answer: `tail Documents/access.log`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/access.log`

Hint 1 (free): `tail` prints the bottom of a file, the opposite end from `head`.

Hint 2 (–5 XP): Run `tail Documents/access.log`, then submit the flag on the last line.

Don't Trust Extensions

20 pts · act2-file · OK

Task: Never trust a filename — an analyst identifies a file by its contents. Determine the true format of `evidence/mystery_file`.

Answer: `file evidence/mystery_file`

Completion: Auto-completes on: commandMatches

Hint 1 (free): One command reads a file's magic bytes and reports what it really is.

Hint 2 (–5 XP): Run `file evidence/mystery_file`.

Also accepted: `file ./evidence/mystery_file`, `file /home/analyst/evidence/mystery_file`

Carving Binary Artifacts

25 pts · act2-strings · OK

Task: `evidence/binary_data` is a compiled executable, so `cat` gives you garbage. Pull the readable text out of it and submit the flag hidden inside.

Answer: `strings evidence/binary_data`

Completion: Submits a flag · flag is inside `/home/analyst/evidence/binary_data`

Hint 1 (free): One command extracts printable character sequences from a binary.

Hint 2 (–5 XP): Run `strings evidence/binary_data`, then submit the recovery token in the output.

Integrity Checksum (MD5)

25 pts · act2-md5 · OK

Task: Digital evidence must be hashed to prove nobody altered it. Compute the MD5 hash of `evidence/evidence.img`, then read the image header to find and submit the acquisition flag.

Answer: `md5sum evidence/evidence.img` **then** `cat evidence/evidence.img`

Completion: Submits a flag · flag is inside `/home/analyst/evidence/evidence.img`

Hint 1 (free): One command computes an MD5 digest. Reading the header afterwards is a plain `cat`.

Hint 2 (–5 XP): Run `md5sum evidence/evidence.img`, then `cat evidence/evidence.img`.

Instructor note: The hash is NOT the flag — the flag is in the file text.

Act III: Search & Discovery

Pattern search, find, manual pages, and the WSL /mnt/c bridge

Secrets in Plain Sight

25 pts · act3-grep · OK

Task: `Documents/secrets.txt` is long, and only one line matters. Find the line containing `vault_passcode` and submit the flag on it.

Answer: `grep vault_passcode Documents/secrets.txt`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/secrets.txt`

Hint 1 (free): `grep` searches a file for a pattern. Give it the pattern first, then the file.

Hint 2 (–5 XP): Run `grep vault_passcode Documents/secrets.txt`.

Case-Insensitive Searching

30 pts · act3-grepi · OK

Task: `Documents/logs.txt` records errors inconsistently as 'error', 'ERROR' and 'Error'. Find every error line with a single search, then submit the flag one of them carries.

Answer: `grep -i "error" Documents/logs.txt`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/logs.txt`

Hint 1 (free): One short `grep` flag makes the match ignore letter case.

Hint 2 (–10 XP): Run `grep -i "error" Documents/logs.txt`.

Needle in /var/log

30 pts · act3-find · OK

Task: Log files are scattered somewhere under `/var/log`. List every file there whose name ends in `.log`, then read the sensor audit log and submit the keycode flag.

Answer: `find /var/log -name "*.log"` **then** `cat /var/log/sensor_audit.log`

Completion: Submits a flag · flag is inside `/var/log/sensor_audit.log`

Hint 1 (free): `find` walks a directory tree, and it can filter on the filename. Quote the wildcard pattern.

Hint 2 (–10 XP): Run `find /var/log -name "*.log"`, then `cat /var/log/sensor_audit.log`.

The WSL Bridge

40 pts · act3-crossing · OK

Task: This machine has a Windows side mounted at `/mnt/c`. Read the intake sheet at `/mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt` and submit the flag.

Answer: `cat /mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt`

Hint 1 (free): WSL mounts each Windows drive under `/mnt`, so drive C: appears as `/mnt/c`.

Hint 2 (–10 XP): Run `cat /mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt`.

The Bridge, Unassisted

45 pts · act3-crossing-solo · OK

Task: A second dossier sits on the Windows side at `C:\Users\analyst\Documents\surface_notes.txt`. Reach it from this Linux shell and submit the flag inside. Nobody will translate the path for you.

Answer: `cat /mnt/c/Users/analyst/Documents/surface_notes.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/analyst/Documents/surface_notes.txt`

Hint 1 (free): Translate the Windows path yourself: `C:\` becomes `/mnt/c/`, and every backslash becomes a forward slash.

Hint 2 (–15 XP): Run `cat /mnt/c/Users/analyst/Documents/surface_notes.txt`.

Instructor note: Deliberately unassisted: the student must translate the Windows path themselves.

Consult the Manual

30 pts · act3-man · OK

Task: The `tracker` sensor suite has an undocumented manual override code, but it is written in the tool's own manual page. Read the manual, find the code, and submit it.

Answer: `man tracker`

Completion: Submits a flag

Hint 1 (free): Every serious Linux tool ships a manual page. One command opens it.

Hint 2 (–10 XP): Run `man tracker` and read the DESCRIPTION section.

Instructor note: Flag is inside the DESCRIPTION section.

Deploy the Sensor Suite (apt-get)

35 pts · act3-apt · OK

Task: The `tracker` tool is not installed on this machine. Install it from the package repository, then run a sensor sweep with it and submit the flag.

Answer: `sudo apt-get update && sudo apt-get install tracker -y` **then** `tracker -a`

Completion: Submits a flag · flag is inside `tracker`

Hint 1 (free): Refresh the package lists first, then install. Both steps need administrator rights. The sweep flag for `tracker` is `-a`.

Hint 2 (–10 XP): Run `sudo apt-get update && sudo apt-get install tracker -y`, then `tracker -a`.

Act IV: The Plumbing

Pipes, redirection, filters, and multi-stage analysis

Filter the Noise (Inverted Grep)

40 pts · act4-grep-v · OK

Task: `Documents/network_stream.log` is almost entirely normal `ALLOW` traffic. Show only the packets that are NOT allowed, and submit the flag on the CRITICAL_DATA_LEAK entry.

Answer: `grep -v "ALLOW" Documents/network_stream.log`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/network_stream.log`

Hint 1 (free): `grep` normally keeps the matching lines. One flag inverts that and keeps everything else.

Hint 2 (–10 XP): Run `grep -v "ALLOW" Documents/network_stream.log`.

First Pipeline (grep | wc -l)

45 pts · act4-pipe-count · OK

Task: You now know how to isolate the non-`ALLOW` packets in `Documents/network_stream.log`. This time do not read them — count them, in a single line, without writing anything to a file.

Answer: `wc -l` **then** `grep -v "ALLOW" Documents/network_stream.log | wc -l`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The `|` operator feeds one command's output straight into the next. `wc -l` counts lines.

Hint 2 (–10 XP): Run `grep -v "ALLOW" Documents/network_stream.log | wc -l`.

Also accepted: `grep -v "ALLOW" Documents/network_stream.log | wc -l`

Extracting Fields (cut & grep)

45 pts · act4-pipe-csv · OK

Task: `Documents/security_events.csv` is comma separated. The rows tagged `FLAG_EMIT` carry a token in column 6. Isolate those rows, extract just that column, and submit the token.

Answer: `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/security_events.csv`

Hint 1 (free): Two commands joined by a pipe: `grep` selects the rows, `cut` selects the column. `cut` needs the delimiter and the field number.

Hint 2 (–10 XP): Run `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6`.

Output Redirection (>)

50 pts · act4-redirect · OK

Task: Pull every error line out of `Documents/logs.txt` and save them into a new file at `/tmp/errors.log` instead of printing them. Then read that file back to confirm it worked.

Answer: `grep -i "error" Documents/logs.txt > /tmp/errors.log` **then** `cat /tmp/errors.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `>` sends output into a file rather than the screen. Remember the flag that makes the search ignore case.

Hint 2 (–10 XP): Run `grep -i "error" Documents/logs.txt > /tmp/errors.log`, then `cat /tmp/errors.log`.

Also accepted: `grep -i "error" Documents/logs.txt > /tmp/errors.log`

Instructor note: The redirect prints nothing — that is correct.

Act V: The Capstone

Multi-step investigation: partition offsets and evidence carving

Partition Geometry Inspection

60 pts · act5-scan · OK

Task: `evidence/suspect_drive.raw` is a raw disk image. Analyze its partition table with the `scan` tool and note the starting sector offset of the Evidence Vault partition. You will need that number next.

Answer: `scan evidence/suspect_drive.raw`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `scan` takes the path to the image.

Hint 2 (–10 XP): Run `scan evidence/suspect_drive.raw` and read the Start Sector Offset it reports.

Also accepted: `scan ./evidence/suspect_drive.raw`

Instructor note: Students must READ the Start sector offset from the table; the capstone no longer prints it for them.

Carve the Evidence (Finish Line)

120 pts · act5-capstone · OK

Task: Carve the Evidence Vault partition out of `evidence/suspect_drive.raw` with the `extract` tool. It needs the sector offset you recovered in the partition scan — go back and read it if you did not write it down. Submit the Master Capstone Flag it decrypts.

Answer: `extract -o 206848 evidence/suspect_drive.raw`

Completion: Submits a flag · flag is inside `act5-capstone`

Hint 1 (free): `extract` takes the offset with the `-o` flag, then the image path. The offset came from the `scan` output; the tool rejects a wrong one.

Hint 2 (–20 XP): Run `extract -o 206848 evidence/suspect_drive.raw`, then submit the master flag it prints.

Instructor note: Chains off act5-scan. A wrong offset is rejected by the tool, so a student who skipped the scan must go back.

Topside (Windows Quest)

Windows CMD parity: dir /a, type, findstr, attrib, and certutil

Windows Navigation (dir & cd)

15 pts · topside-nav · OK

Task: You are now on the Windows side. List the files in `C:\Users\Analyst` using the Windows equivalent of `ls`.

Answer: `dir`

Completion: Auto-completes on: commandMatches

Hint 1 (free): In cmd.exe the listing command is `dir`. On its own, `cd` prints where you are.

Hint 2 (–5 XP): Run `dir`.

Also accepted: `dir .`

Hidden File Attributes (attrib)

20 pts · topside-attrib · OK

Task: Windows hides files with an attribute flag, not with a leading dot. Inspect the attributes of `evidence\mystery_file`, then read it and recover the flag.

Answer: `attrib evidence\mystery_file` **then** `type evidence\mystery_file`

Completion: Submits a flag · flag is inside `C:\Users\Analyst\evidence\mystery_file`

Hint 1 (free): `attrib` shows a file's attribute letters; H means Hidden. `type` prints a file, the way `cat` does on Linux.

Hint 2 (–5 XP): Run `attrib evidence\mystery_file`, then `type evidence\mystery_file`.

Windows String Search (findstr /i)

25 pts · topside-findstr · OK

Task: Search the Windows event log at `Documents\logs.txt` for the flagged event containing `marker`, ignoring letter case, and submit the flag. Windows does not have `grep`.

Answer: `findstr /i "marker" Documents\logs.txt`

Completion: Submits a flag · flag is inside `C:\Users\Analyst\Documents\logs.txt`

Hint 1 (free): The cmd.exe search command is `findstr`. Its switches start with a forward slash, and `/i` ignores case.

Hint 2 (–5 XP): Run `findstr /i "marker" Documents\logs.txt`.

CertUtil Hashing

20 pts · topside-certutil · OK

Task: Compute the MD5 hash of `evidence\evidence.img` for the chain of custody. Windows has no `md5sum`; the built-in tool for this is `certutil`.

Answer: `certutil -hashfile evidence\evidence.img MD5`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `certutil` needs the `-hashfile` switch, then the file, then the algorithm name.

Hint 2 (–5 XP): Run `certutil -hashfile evidence\evidence.img MD5`.

Also accepted: `certutil -hashfile evidence/evidence.img MD5`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes around a path
- Path prefixes: `./Documents`, `~/Documents`, or the full absolute path
- Short paths when already inside the folder
- Flag spacing: `head -n 5`, `head -n5`, `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent. Point stuck students there before they buy a hint.
- **Unsimulated tools** reply with what they are, rather than a bare "command not found".
- **Hints:** the first is free and conceptual; the one that gives the exact command costs XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.
- **Watch the stuck points.** The dashboard flags any challenge with a solve rate under 35%. Now that briefs no longer contain the answer, that number means something: it marks a concept the class did not get, not a typing error.
